

Chi Hang Chan

associate professor

基本信息

姓名: Chi Hang Chan

所属院校: 澳门大学

地区: China-Macau

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

混合信号集成技术

模拟与射频IC设计

低功耗电路与能量管理

个人简介

Chi Hang Chan, Ph.D., is an Associate Professor at the State-Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), University of Macau. He holds a Ph.D. in Electrical and Computer Engineering from the University of Macau and a B.Sc. from the University of Washington. His research focuses on analog and mixed-signal CMOS integrated circuits, high-speed Nyquist ADCs, wideband SDMs, low-jitter ring-VCO-based PLLs, and PUFs. Dr. Chan teaches courses in analog integrated circuit design and research methodology at both undergraduate and graduate levels. He has supervised numerous theses and serves as a reviewer for several prestigious journals.

工作经历

State-Key Laboratory of Analog and Mixed-Signal VLSI (AMSV),
University of Macau - Associate Professor

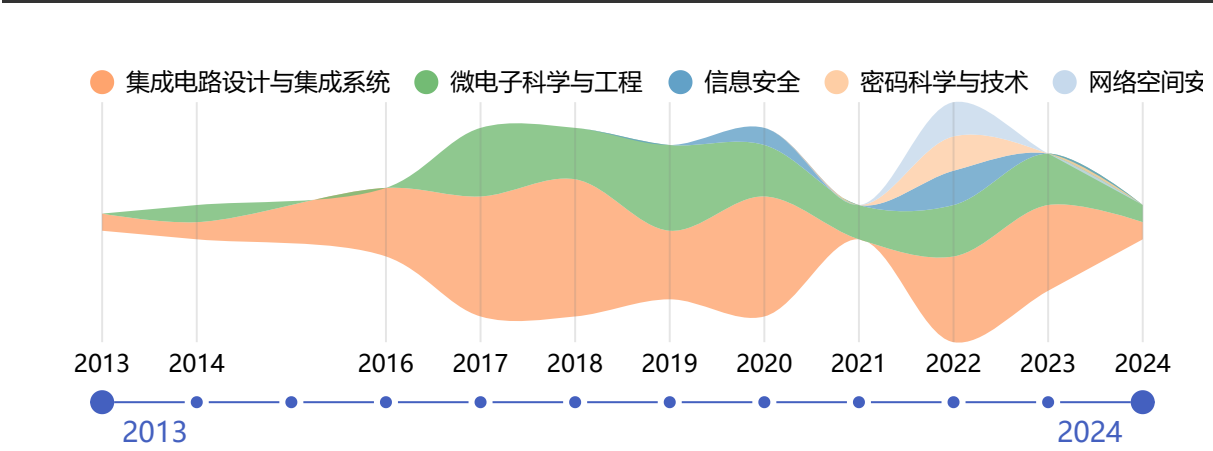
2022-08 - 至今

State-Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), University of Macau - Assistant Professor	2017-04 - 2022-07
State-Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), University of Macau - Research Assistant Professor	2016-01 - 2017-03
State-Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), University of Macau - Post-doc Fellow	2015-08 - 2015-12
Dept. of EEE, University of California - Special Scientist	2016-03 - 2016-07

教育经历

University of Macau - Ph.D.	2011 - 2015
University of Macau - M.Sc.	- 2011
University of Washington - B.Sc.	- 2008

近年发表的论文



An ELDC-Free 4th-Order CT SDM Facilitated by 2nd-Order NS CT-SAR and AC-Coupled Negative-R

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xu, Zixuan; Xing, Kai; Zhu, Yan; Martins, Rui P.; Chan, Chi-Hang

被引用次数: 1

2024

A 0.016mm² Active Area 4GHz Fully Ring-Oscillator-Based Cascaded Fractional-N PLL With Burst-Mode Sampling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhong, Junlin; Yang, Xiaofeng; Martins, Rui P.; Zhu, Yan; Chan, Chi-Hang

被引用次数: 0

2023

A 12-bit 1GS/s ADC With Background Distortion and Split-ADC-Like Gain Calibration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wei, Lai; Zheng, Zihao; Markulic, Nereo; Lagos, Jorge; Martens, Ewout; Martins, Rui Paulo; Zhu, Yan; Craninckx, Jan; Chan, Chi-Hang

被引用次数: 3

2023

A 52.5-dB 2x Time-Interleaved 2.8-GS/s SAR ADC With 5-bit/Cycle Time-Domain Quantization and a Compact Signal DAC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhao, Hongzhi; Zhang, Minglei; Zhu, Yan; Martins, Rui P.; Chan, Chi-Hang

被引用次数: 1

2023

A Second-Order NS Pipelined SAR ADC With Quantization-Prediction-Unrolled Gain Error Shaping and Fully Passive Integrator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Hongshuai; Zhu, Yan; Martins, Rui P.; Chan, Chi-Hang

被引用次数: 2

2023

A 14b 500 MS/s Single-Channel Pipelined-SAR ADC With Reference Ripple Mitigation Techniques and Adaptively Biased Floating Inverter Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Wenning; Zhu, Yan; Chen, Chixiao; Xu, Hao; Liu, Qi; Liu, Ming; Martins, Rui P.; Chan, Chi-Hang

被引用次数: 4

2023

A Weak PUF-Assisted Strong PUF With Inherent Immunity to Modeling Attacks and Ultra-Low BER

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Jiahao; Zhao, Yuanzhe; Zhu, Yan; Chan, Chi-Hang; Martins, Rui Paulo

被引用次数: 14

2022

An Entropy-Source-Preselection-Based Strong PUF With Strong Resilience to Machine Learning Attacks and High Energy Efficiency

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Jiahao; Zhu, Yan; Chan, Chi-Hang; Martins, Rui Paulo

被引用次数: 1

2022

Supply-Noise-Desensitized Techniques for Low Jitter RO-Based PLL Achieving ≤ 1.6 ps RMS Jitter Within Full-Spectrum Supply Interference

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Duan, Yu; Chan, Chi-Hang; Zhu, Yan; Martins, Rui Paulo

被引用次数: 1

2022

A Single-Opamp Third Order CT $\Delta\sigma$ Modulator With SAB-ELD-Merged Integrator and Three-Stage Hybrid Compensation Opamp

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xing, Kai; Wang, Wei; Zhu, Yan; Chan, Chi-Hang; Martins, Rui P.

被引用次数: 5

2022

A 3.3-GS/s 6-b Fully Dynamic Pipelined ADC With Linearized Dynamic Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zheng, Zihao; Wei, Lai; Lagos, Jorge; Martens, Ewout; Zhu, Yan; Chan, Chi-Hang; Craninckx, Jan; Martins, Rui P.

被引用次数: 9

2022

An Inherent Gain Error Tolerance Noise-Shaping SAR-Assisted Pipeline ADC With Code-Counter-Based Offset Calibration

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Hongshuai; Zhu, Yan; Chan, Chi-Hang; Martins, Rui P.

被引用次数: 8

2022

A 20 MHz Bandwidth 79 dB SNDR SAR-Assisted Noise-Shaping Pipeline ADC With Gain and Offset Calibrations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Yanbo; Zhang, Jin; Liu, Shubin; Ding, Ruixue; Zhu, Yan; Chan, Chi-Hang; Martins, Rui P.

被引用次数: 6

2022

A 40-MHz Bandwidth 75-dB SNDR Partial-Interleaving SAR-Assisted Noise-Shaping Pipeline ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Yan; Zhu, Yan; Chan, Chi-Hang; Martins, Rui P.

被引用次数: 15

2021

A 7-bit 2 GS/s Time-Interleaved SAR ADC With Timing Skew Calibration Based on Current Integrating Sampler

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jiang, Wenning; Zhu, Yan; Chan, Chi-Hang; Murmann, Boris; Martins, Rui Paulo

被引用次数: 25

2021

An 8-Bit 10-GS/s 16x Interpolation-Based Time-Domain ADC With <1.5-ps Uncalibrated Quantization Steps

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Minglei; Zhu, Yan; Chan, Chi-Hang; Martins, Rui P.

被引用次数: 35

2020

A 100-MHz BW 72.6-dB-SNDR CT $\Delta\sigma$ Modulator Utilizing Preliminary Sampling and Quantization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Wei; Chan, Chi-Hang; Zhu, Yan; Martins, Rui P.

被引用次数: 13

2020

A 12.5-MHz Bandwidth 77-dB SNDR SAR-Assisted Noise Shaping Pipeline ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Yan; Chan, Chi-Hang; Zhu, Yan; Martins, Rui P.

被引用次数: 18

2020

A Temperature-Stabilized Single-Channel 1-GS/s 60-dB SNDR SAR-Assisted Pipelined ADC With Dynamic Gm-R-Based Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Wenning; Zhu, Yan; Zhang, Minglei; Chan, Chi-Hang; Martins, Rui Paulo

被引用次数: 46

2020

A 0.04% BER Strong PUF With Cell-Bias-Based CRPs Filtering and Background Offset Calibration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Jiahao; Zhu, Yan; Chan, Chi-Hang; Martins, Rui Paulo

被引用次数: 8

2020

A Calibration-Free Ring-Oscillator PLL With Gain Tracking Achieving 9% Jitter Variation Over PVT

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yang, Xiaofeng; Chan, Chi-Hang; Zhu, Yan; Martins, Rui Paulo

被引用次数: 8

2020

An 11-bit 100-MS/s Pipelined-SAR ADC Reusing PVT-Stabilized Dynamic Comparator in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Jin; Ren, Xiaoqian; Liu, Shubin; Chan, Chi-Hang; Zhu, Zhangming

被引用次数: 9

2020

A 2nd-Order Noise-Shaping SAR ADC With Lossless Dynamic Amplifier Assisted Integrator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Yanbo; Liu, Shubin; Tian, Binbin; Zhu, Yan; Chan, Chi-Hang; Zhu, Zhangming

被引用次数: 26

2020

Design of a High-Speed Time-Interleaved Sub-Ranging SAR ADC With Optimal Code Transfer Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xing, Dezhi; Zhu, Yan; Chan, Chi-Hang; Maloberti, Franco; Seng-Pan, U.; Martins, Rui Paulo

被引用次数: 2

2019

A 4-b 7- μ W Phase Domain ADC With Time Domain Reference Generation for Low-Power FSK/PSK Demodulation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lei, Xuewei; Chan, Chi-Hang; Zhu, Yan; Martins, Rui Paulo

被引用次数: 0

2019

Analysis of Reference Error in High-Speed SAR ADCs With Capacitive DAC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Li, Cheng; Chan, Chi-Hang; Zhu, Yan; Martins, Rui P.

被引用次数: 12

2019

Background Offset Calibration for Comparator Based on Temperature Drift Profile

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Li, Xiaochao; Chan, Chi-Hang; Zhang, Qi; Zhu, Yan; Martins, R. P.

被引用次数: 3

2019

A 0.6-V 13-bit 20-MS/s Two-Step TDC-Assisted SAR ADC With PVT Tracking and Speed-Enhanced Techniques

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Minglei; Chan, Chi-Hang; Zhu, Yan; Martins, Rui P.

被引用次数: 35

2019

Gain Error Calibrations for Two-Step ADCs: Optimizations Either in Accuracy or Chip Area

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wang, Guan Cheng; Zhu, Yan; Chan, Chi-Hang; Seng-Pan, U.; Martins, Rui P.

被引用次数: 2

2018

A Two-Way Interleaved 7-b 2.4-GS/s 1-Then-2 b/Cycle SAR ADC With Background Offset Calibration

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chan, Chi-Hang; Zhu, Yan; Zhang, Wai-Hong; Seng-Pan, U.; Martins, Rui Paulo

被引用次数: 67

2018

A 5.35-mW 10-MHz Single-Opamp Third-Order CT $\Delta \Sigma$ Modulator With CTC Amplifier and Adaptive Latch DAC Driver in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Wei; Zhu, Yan; Chan, Chi-Hang; Martins, Rui Paulo

被引用次数: 13

2018

Passive Noise Shaping in SAR ADC With Improved Efficiency

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Song, Yan; Chan, Chi-Hang; Zhu, Yan; Geng, Li; Seng-Pan, U.; Martins, Rui Paulo

被引用次数: 23

2018

Missing-Code-Occurrence Probability Calibration Technique for DAC Nonlinearity With Supply and Reference Circuit Analysis in a SAR ADC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Guancheng; Li, Cheng; Zhu, Yan; Zhong, Jianyu; Lu, Yan; Chan, Chi-Hang; Martins, Rui P.

被引用次数: 5

2018

A 0.19 mm² 10 b 2.3 GS/s 12-Way Time-Interleaved Pipelined-SAR ADC in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhu, Yan; Chan, Chi-Hang; Zheng, Zi-Hao; Li, Cheng; Zhong, Jian-Yu; Martins, Rui P.

被引用次数: 14

2018

Analysis of Common-Mode Interference and Jitter of Clock Receiver Circuits With Improved Topology

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yang, Xiaofeng; Zhu, Yan; Chan, Chi-Hang; Seng-Pan, U.; Martins, Rui P.

被引用次数: 8

2018

A 2-GS/s 8-Bit ADC Featuring Virtual-Ground Sampling Interleaved Architecture in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Wang, X. Shawn; Jin, Xin; Du, Jieqiong; Li, Yilei; Du, Yuan; Wong, Chien-Heng; Kuan, Yen-Cheng; Chan, Chi-Hang; Chang, Mau-Chung Frank

被引用次数: 5

2018

Metastability in SAR ADCs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chan, Chi-Hang; Zhu, Yan; Sin, Sai-Weng; Murmann, Boris; Seng-Pan, U.; Martins, R. P.

被引用次数: 15

2017

A 10-bit 500-MS/s Partial-Interleaving Pipelined SAR ADC With Offset and Reference Mismatch Calibrations

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhu, Yan; Chan, Chi-Hang; Pan, Seng U.; Martins, Rui Paulo

被引用次数: 11

2017

Active-Passive $\Delta \Sigma$ Modulator for High-Resolution and Low-Power Applications

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hussain, Arshad; Sin, Sai-Weng; Chan, Chi-Hang; U, Seng-Pan (Ben); Maloberti, Franco; Martins, Rui P.

被引用次数: 23

2017

Seven-bit 700-MS/s Four-Way Time-Interleaved SAR ADC With Partial Vcm-Based Switching

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Xing, Dezhi; Zhu, Yan; Chan, Chi-Hang; Sin, Sai-Weng; Ye, Fan; Ren, Junyan; U, Seng-Pan; Martins, Rui Paulo

被引用次数: 15

2017

A 12b 180MS/s 0.068mm² With Full-Calibration-Integrated Pipelined-SAR ADC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhong, Jianyu; Zhu, Yan; Chan, Chi-Hang; Sin, Sai-Weng; U, Seng-Pan; Martins, Rui Paulo

被引用次数: 25

2017

A 7.8-mW 5-b 5-GS/s Dual-Edges-Triggered Time-Based Flash ADC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chan, Chi-Hang; Zhu, Yan; Sin, Sai-Weng; Seng-Pan, U.; Martins, Rui P.; Maloberti, Franco

被引用次数: 28

2017

60-dB SNDR 100-MS/s SAR ADCs With Threshold Reconfigurable Reference Error Calibration

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chan, Chi-Hang; Zhu, Yan; Li, Cheng; Zhang, Wai-Hong; Ho, Iok-Meng; Wei, Lai; Seng-Pan, U.; Martins, Rui Paulo

被引用次数: 39

2017

A 6 b 5 GS/s 4 Interleaved 3 b/Cycle SAR ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chan, Chi-Hang; Zhu, Yan; Sin, Sai-Weng; Seng-Pan, (Ben) U.; Martins, Rui Paulo

被引用次数: 36

2016

An 11b 450 MS/s Three-Way Time-Interleaved Subranging Pipelined-SAR ADC in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhu, Yan; Chan, Chi-Hang; U, Seng-Pan; Martins, Rui Paulo

被引用次数: 32

2016

Uniform Quantization Theory-Based Linearity Calibration for Split Capacitive DAC in an SAR ADC

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Liu, Jianwei; Zhu, Yan; Chan, Chi-Hang; Sin, Sai-Weng; Seng-Pan, U.; Martins, Rui Paulo da Silva

被引用次数: 14

2016

Histogram-Based Ratio Mismatch Calibration for Bridge-DAC in 12-bit 120 MS/s SAR ADC

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhu, Yan; Chan, Chi-Hang; Wong, Si-Seng; Seng-Pan, U.; Martins, Rui Paulo

被引用次数: 29

2016

Split-SAR ADCs: Improved Linearity With Power and Speed Optimization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhu, Yan; Chan, Chi-Hang; Chio, U-Fat; Sin, Sai-Weng; U, Seng-Pan; Martins, Rui Paulo; Maloberti, Franco

被引用次数: 49

2014

A 5-Bit 1.25-GS/s 4x-Capacitive-Folding Flash ADC in 65-nm CMOS

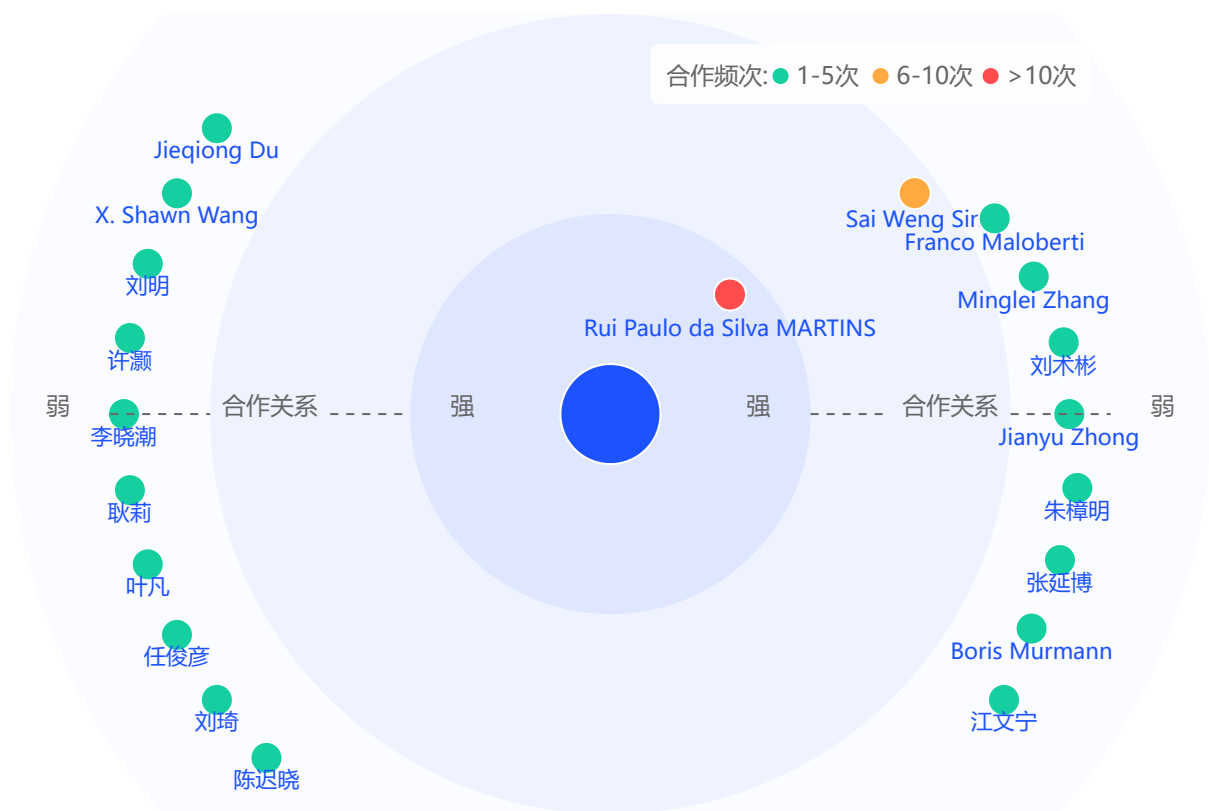
IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chan, Chi-Hang; Zhu, Yan; Sin, Sai-Weng; U, Seng-Pan; Martins, Rui Paulo; Maloberti, Franco

被引用次数: 20

2013

学术合作



Rui Paulo da Silva MARTINS

合作次数: 24

澳门大学

Sai Weng Sin

合作次数: 9

澳门大学

Franco Maloberti

合作次数: 5

帕维亚大学

Minglei Zhang

合作次数: 4

澳门大学

刘术彬

合作次数: 3

西安电子科技大学

还有 17 个合作者